

2.3 Curves in the Complex Plane

2 Complex roots of $f(z)$

Question 1.

First, we observe that

$$f_1(z) = 0 \iff z^3 = z^2 - (2 - i)z + 1 + i$$

Also

$$\begin{aligned} |z| > 3 &\implies |z|^3 > 3|z|^2 > |z|^2 + 3|z| + 9 > |z^2 - (2 - i)z + 1 + i| \\ &\implies z \text{ is not a root of } f_1 \end{aligned}$$

So we see that the roots of f_1 lie in $|z| < 3$.

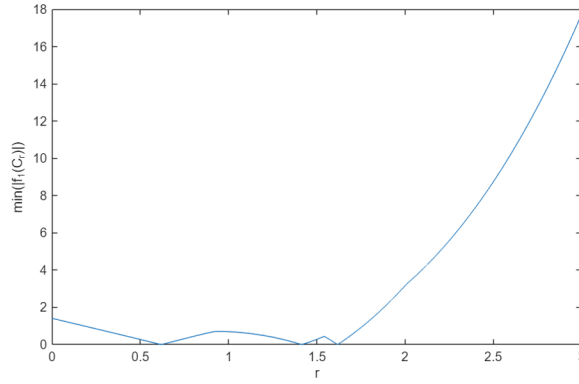


Figure 1: $\min |f_1(C_r)|$ against r for $0 \leq r \leq 3$

We plot the graph of $\min |f_1(C_r)|$ against r for $0 \leq r \leq 3$ (where, for each value of r , the minimum is taken over $\{z = r \exp(\frac{2n\pi i}{25000}) : n = 1, \dots, 25000\} \subseteq C_r$). From Figure (1), we see that the three roots z_1, z_2, z_3 have $r_1 \triangleq |z_1| \approx 0.6$, $r_2 \triangleq |z_2| \approx 1.4$, $r_3 \triangleq |z_3| \approx 1.6$.

We run our program again for the intervals $0.55 \leq r_1 \leq 0.65$, $1.35 \leq r_2 \leq 1.45$ and $1.55 \leq r_3 \leq 1.65$ (where we have calculated $\min |f_1(C_r)|$ for 25000 linearly spaced values in each interval) to find $\arg \min |f_1(C_r)|$ in each interval. This gives us the following results for the values of r_1, r_2, r_3 and the corresponding computed roots $\hat{z}_1, \hat{z}_2, \hat{z}_3$.

i	$\hat{r}_i$	$\hat{z}_i$	$\hat{z}_i$ (3s.f.)
1	0.618035	$-0.0000388338 + 0.618035i$	$0.000 + 0.618i$
2	1.41421	$0.999969 + 1.00003i$	$1.00 + 1.00i$
3	1.61803	$-0.000101668 - 1.61803i$	$0.00 - 1.62i$

Table 1: Computed values for the roots of f_1

We observe that

$$\begin{aligned} f_1(z) &= (z - 1 - i)(z^2 + iz + 1) \\ &= (z - 1 - i) \left(z + \frac{1 - \sqrt{5}}{2}i \right) \left(z + \frac{1 + \sqrt{5}}{2}i \right) \end{aligned}$$

so the roots of $f_1(z)$ are exactly given by

$$\begin{aligned} z_1 &= \frac{1 - \sqrt{5}}{2}i = 0.618034\dots i \approx 0.618i \\ z_2 &= 1 + i \\ z_3 &= -\frac{1 + \sqrt{5}}{2}i = -1.618034\dots i \approx -1.62i \end{aligned}$$

Thus our computed values $\hat{z}_1, \hat{z}_2, \hat{z}_3$ are correct to three significant figures.

Question 2.

We have

$$f'_1(w) = 3w^2 - 2w + 2 - i$$

We see that

$$\begin{aligned} |w| > 2 &\implies 3|w|^2 > 2|w| + 4 > |2w - 2 + i| \\ &\implies z \text{ is not a root of } f'_1 \end{aligned}$$

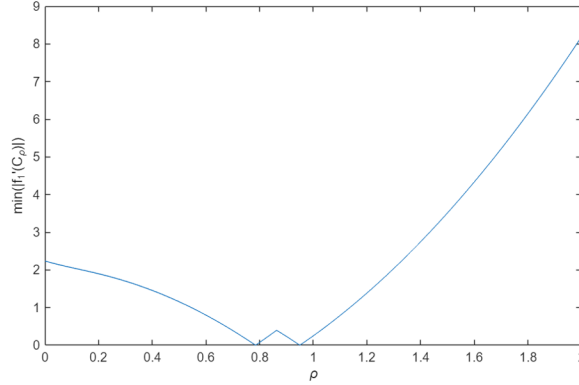


Figure 2: $\min |f'_1(C_\rho)|$ against ρ for $0 \leq \rho \leq 2$

As before, plotting the graph of $\min |f'_1(C_\rho)|$ against ρ for $0 \leq \rho \leq 2$, we see that the two roots have $\rho_1 \triangleq |w_1| \approx 0.8$ and $\rho_2 \triangleq |w_2| \approx 0.95$.

Running our program again for the intervals $0.75 \leq |\rho_1| \leq 0.85$ and $0.90 \leq |\rho_2| \leq 1.00$ gives us the following results for the values of ρ_1, ρ_2 and the corresponding computed roots $\hat{w}_1, \hat{w}_2$.

i	$\hat{\rho}_i$	$\hat{w}_i$	$\hat{w}_i$ (3s.f.)
1	0.784701	$0.118423 - 0.775714i$	$0.118 - 0.776i$
2	0.949858	$0.548193 + 0.775703i$	$0.548 + 0.776i$

Table 2: Computed values for the roots of f'_1

Using the quadratic formula, the roots of f'_1 are exactly given by

$$w_{\pm} = \frac{1}{3} \left(1 \pm \sqrt{-5 + 3i} \right)$$

$$w_1 = \left(\frac{1}{3} - \sqrt{\frac{\sqrt{34} - 5}{18}} \right) - \left(\sqrt{\frac{\sqrt{34} + 5}{18}} \right) i \approx 0.118 - 0.776i$$

$$w_2 = \left(\frac{1}{3} + \sqrt{\frac{\sqrt{34} - 5}{18}} \right) + \left(\sqrt{\frac{\sqrt{34} + 5}{18}} \right) i \approx 0.548 + 0.776i$$

Thus our computed values $\hat{w}_1, \hat{w}_2$ are correct to three significant figures.

3 Images $f(C_r)$ of C_r

Question 3.

Firstly, for all values of r , we observe that the curve $g(C_r)$ has rectangular symmetry. For all $w \in g(C_r)$, we have $w = g(z)$ for some $z \in C_r$. Then since $\bar{z}, -\bar{z} \in C_r$, we have $\bar{w} = g(\bar{z}) \in g(C_r)$, which gives rise to the line of symmetry along $\text{Im}(w) = 0$, and $-\bar{w} = -\bar{z}^3 - \bar{z} = g(-\bar{z}) \in g(C_r)$, which gives rise to the line of symmetry along $\text{Re}(w) = 0$.

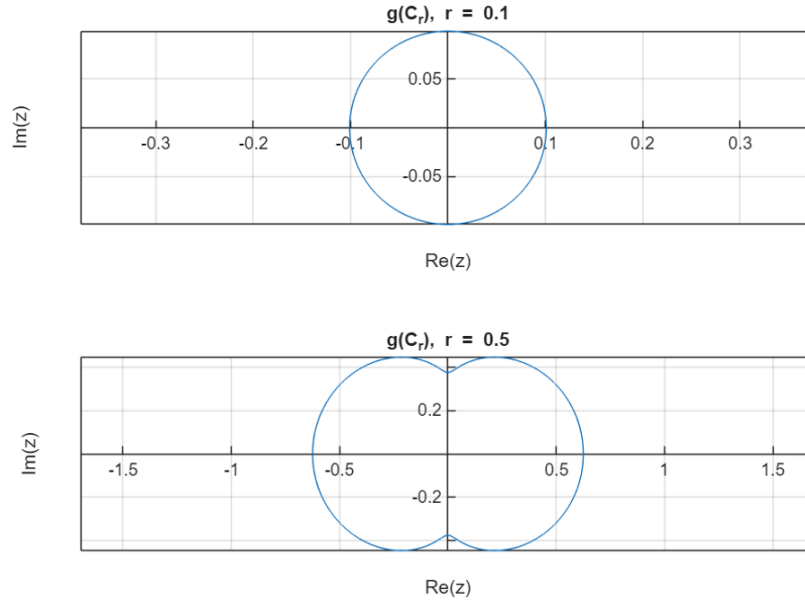


Figure 3: $g(C_r)$ for $r = 0.1$ and $r = 0.5$

For very small values of $r < 1$, we see that $g(C_r)$ approximates the simple circle C_r . At any point $g(z) \in g(C_r)$, the distance from $g(z)$ to $z \in C_r$ is $|g(z) - z| = |z^3| = r^3$, which goes to 0 as $r \rightarrow 0$.

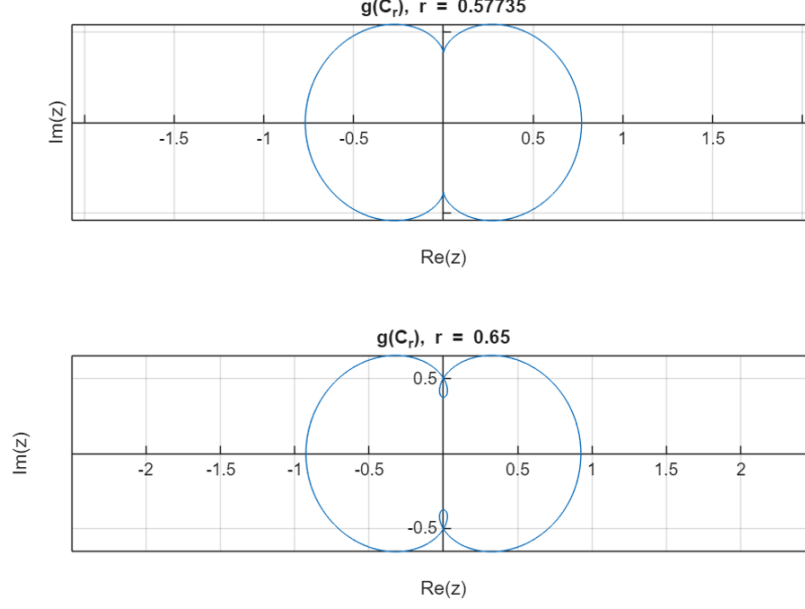


Figure 4: $g(C_r)$ for $r = \frac{1}{\sqrt{3}}$, $r = 0.65$

As r increases from $r = 0$, we see that $g(C_r)$ remains a simple curve as the top and bottom deform inwards, up until $r = r_1^{(g)} \approx 0.577$. For $r > r_1^{(g)}$, we see two self-intersections along the imaginary axis, and two additional loops form inside, within the half-planes $\{\text{Im}(w) > 0\}$ and $\{\text{Im}(w) < 0\}$ for $r < 1$.

To solve for $r_1^{(g)}$, we first note that a self-intersection occurs at $w \in g(C_r)$ if and only if $g(z) = g(z') = w$ for some $z, z' \in C_r$ distinct. Then, by rectangular symmetry, we have self-intersections at $\pm w = g(\pm z) = g(\pm z')$ and $\pm \bar{w} = g(\pm \bar{z}) = g(\pm \bar{z}')$. The fact that there are only two self-intersections, both of which lie along the imaginary axis, tell us that $w = -\bar{w}$, so $g(z) = g(-\bar{z})$. Writing $z = re^{i\phi} \in C_r$, we see that

$$\begin{aligned}
 g(z) = -g(\bar{z}) &\implies z^3 + \bar{z}^3 + z + \bar{z} = 0 \\
 &\implies 2r^3 \cos(3\phi) + 2r \cos \phi = 0 \\
 &\implies r^2(4 \cos^3 \phi - 3 \cos \phi) = -\cos \phi \\
 &\implies \cos \phi = 0 \\
 &\text{or } 4 \cos^2 \phi - 3 = -\frac{1}{r^2} \quad (*)
 \end{aligned}$$

If $\cos \phi = 0$, then z is purely imaginary, so z and $z' = -\bar{z}$ are not distinct, and there are no self-intersections. Also, $(*)$ only has solutions for $-\frac{1}{r^2} \geq 3$, which is true if and only if $r \geq \frac{1}{\sqrt{3}}$ (since $r > 0$). Hence we see that $r_1^{(g)} = \frac{1}{\sqrt{3}}$. At $r = r_1^{(g)}$, we see that the only solutions to $(*)$ have $\cos \phi = 0$, so there are no self-intersections and the curve is simple. For $r > r_1^{(g)}$, we note that there are two self-intersections along the imaginary axis at $w = g(re^{i\phi_0}) = g(re^{i(\pi-\phi_0)})$ and $-w = g(re^{-i\phi_0}) = g(re^{i(\pi+\phi_0)})$, where $\phi_0 \in [0, \frac{\pi}{2}]$ satisfies $(*)$.

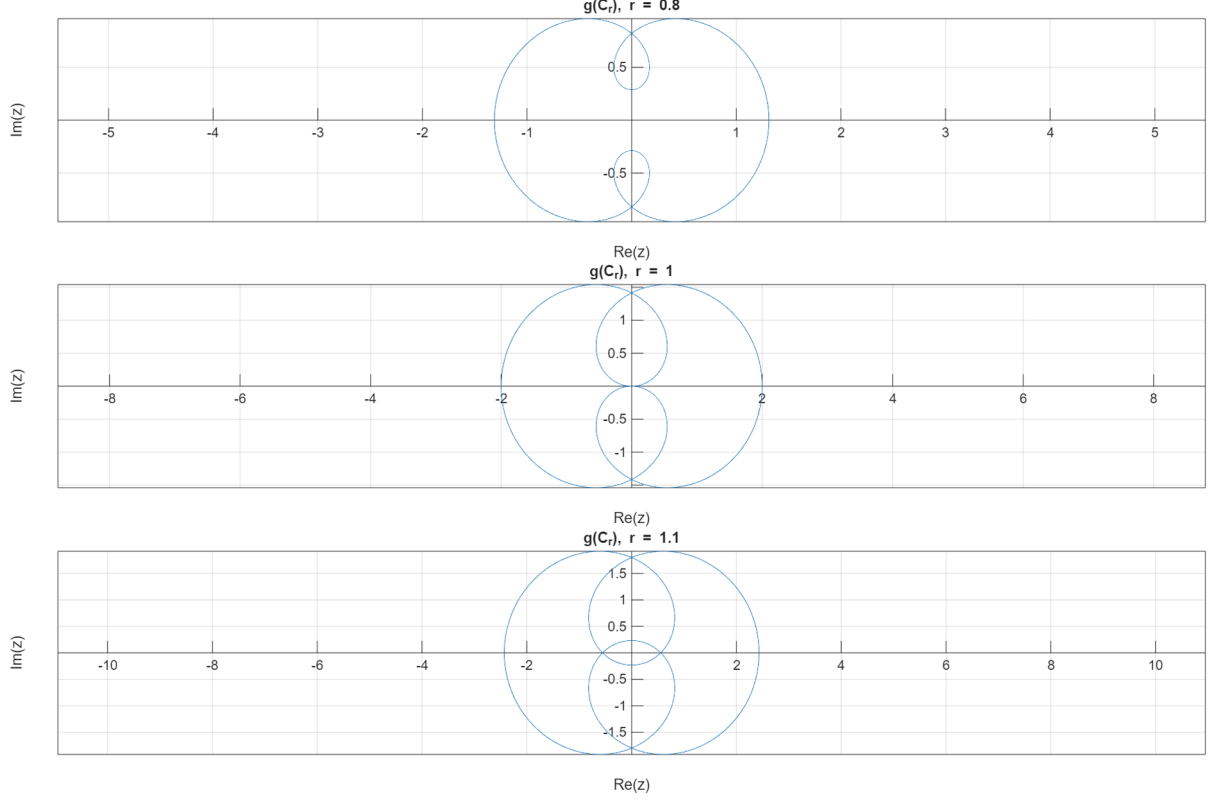


Figure 5: $g(C_r)$ for $r = 0.8$, $r = 1$, $r = 1.1$

As r continues to increase beyond $r = r_1^{(g)}$, we see that the two interior loops increase in size relative to the outer loop. At $r = r_2^{(g)} = 1$, the two loops intersect exactly once at the origin, for a total of 3 self-intersections of $g(C_r)$. For $r > r_2^{(g)}$ the two loops intersect twice along the real axis, resulting in a total of 4 self-intersections of $g(C_r)$.

By the same reasoning as before, the fact that there are only two self-intersections which lie along the positive- and negative-real half-axes tell us that these intersections occur where $w = g(z) = g(\bar{z})$. Writing $z = re^{i\phi} \in C_r$, we see that

$$\begin{aligned}
 g(z) = g(\bar{z}) &\implies z^3 - \bar{z}^3 + z - \bar{z} = 0 \\
 &\implies 2ir^3 \sin(3\phi) + 2ir \sin \phi = 0 \\
 &\implies r^2(-4 \sin^3 \phi + 3 \sin \phi) = -\sin \phi \\
 &\implies \sin \phi = 0 \\
 &\text{or } 4 \sin^2 \phi - 3 = \frac{1}{r^2} \quad (\dagger)
 \end{aligned}$$

If $\sin \phi = 0$, then z is purely real, so z and $z' = \bar{z}$ are not distinct and there are no self-intersections. Also, $(\dagger)$ only has solutions for $\frac{1}{r^2} \leq 1$, which is true if and only if $r \geq 1$, agreeing with our observation that $r_2^{(g)} = 1$. In this case, at $r = r_2^{(g)}$, we see that the solutions have $\sin^2 \phi = 1$, which corresponds to $z = \pm i$, and there is exactly one self-intersection along the real axis. For $r > r_2^{(g)}$, there are two self-intersections along the real axis at $w = g(re^{i\phi_0}) = g(re^{-i\phi_0})$ and $-w = g(re^{i(\pi-\phi_0)}) = g(re^{i(\pi+\phi_0)})$, where $\phi_0 \in [0, \frac{\pi}{2}]$ satisfies $(\dagger)$.

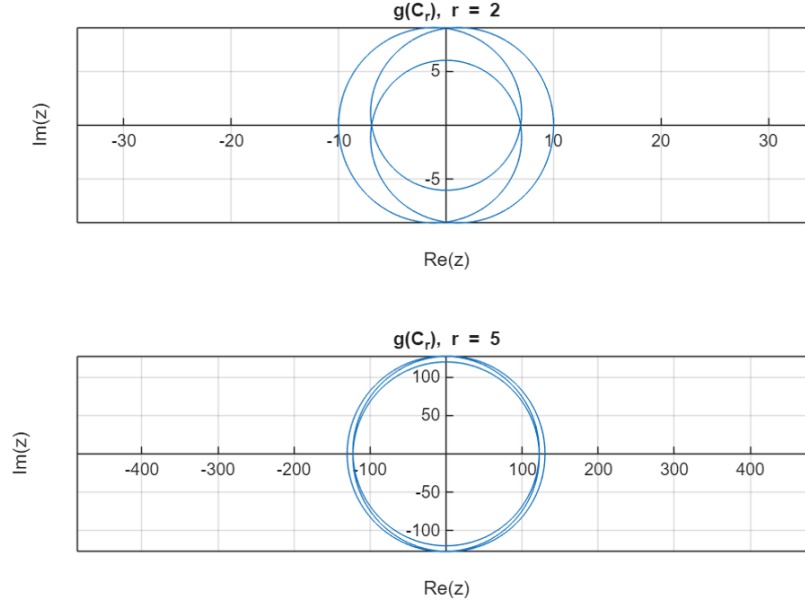


Figure 6: $g(C_r)$ for $r = 2$, $r = 5$

As r increases to ∞ , the two interior loops continue to increase relative in size to the outer loop, so that the three loops become approximately circular and approximately equal in size, centred on the origin. For large r , the z^3 term in $g(z) = z^3 + z$ dominates. The image of C_r under $z \mapsto z^3$ is a circle of radius r^3 which winds around the origin thrice, agreeing with the limiting behaviour of $g(C_r)$ as $r \rightarrow \infty$.

Note that the curve $g(C_r)$ naturally partitions $\mathbb{C} \setminus g(C_r)$ into path components, within which the winding number $I(g(C_r); w)$ is locally constant. Also, observe that for each $w \in \mathbb{C} \setminus g(C_r)$, we have

$$I(g(C_r); w) = |\{z \in B(0; r) : g(z) = w\}|$$

where $B(z; r)$ is the open ball of radius r centred on z . Therefore, the curve $g(C_r)$ can also be thought of as the union of the boundaries of the regions $\{w \in \mathbb{C} : z^3 + z - w \text{ has } n \text{ roots in } B(0; r)\} \subseteq \mathbb{C}$ for $n = 0, 1, 2, 3$.

Question 4.

Firstly, for all values of r , we observe that the curve $h(C_r)$ has symmetry about the real axis. For all $w \in h(C_r)$, we have $w = h(z)$ for some $z \in C_r$. Then since $\bar{z} \in C_r$, we have $\bar{w} = h(\bar{z}) \in h(C_r)$, which gives rise to the line of symmetry along $\text{Im}(w) = 0$.

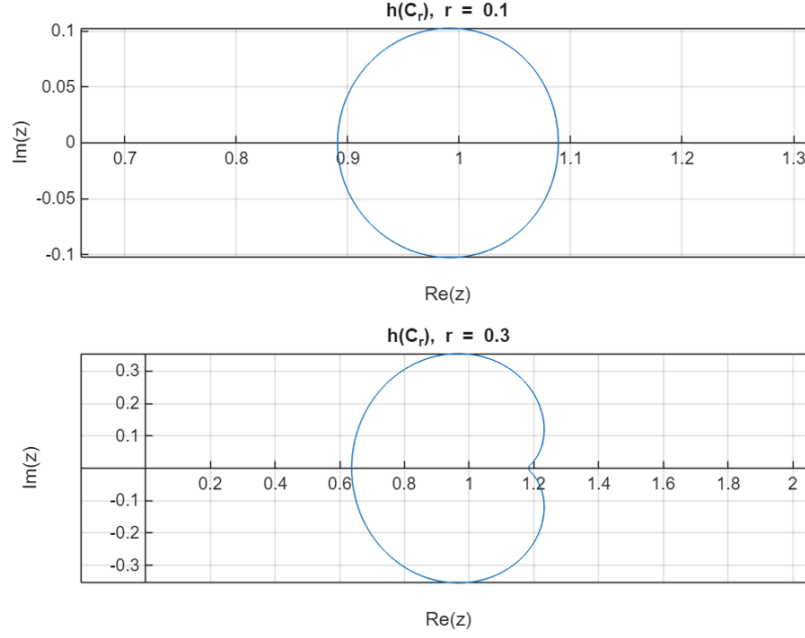


Figure 7: $h(C_r)$ for $r = 0.1$ and $r = 0.3$

For very small values of $r < 1$, we see that $h(C_r)$ is approximately circular. As before, for small r , the $-z + 1$ terms in $h(z) = z^3 - z^2 - z + 1$ dominate, and the image of C_r under $z \mapsto -z + 1$ is a circle of radius r centred on 1, explaining this observation.

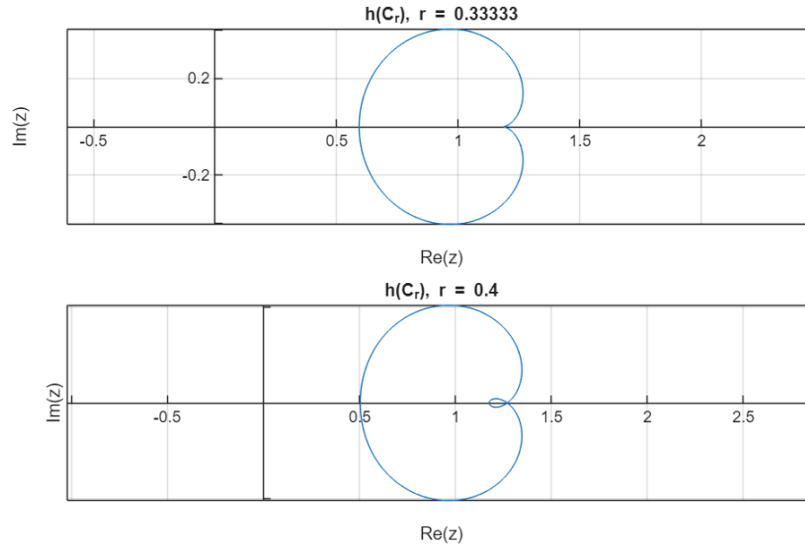


Figure 8: $h(C_r)$ for $r = \frac{1}{3}$, $r = 0.4$

As r increases from $r = 0$, we see that $h(C_r)$ remains a simple curve as the right side deforms inwards, up until $r = r_1^{(h)} \approx 0.33$. For $r > r_1^{(h)}$, we see a self-intersection on the right, and an additional loop forms inside.

To solve for $r_1^{(h)}$, we observe that the self-intersection along the real axis occurs when $w = h(z) = h(\bar{z}) \in \mathbb{R}$

for some $z \in C_r \setminus \mathbb{R}$. Writing $z = re^{i\phi}$, we see that

$$\begin{aligned}
\operatorname{Im}(h(z)) = 0 &\implies r^3 \sin(3\phi) - r^2 \sin(2\phi) - r \sin(\phi) = 0 \\
&\implies r \sin \phi (-4r^2 \sin^2 \phi + 3r^2 - 2r \cos \phi - 1) = 0 \\
&\implies r \sin \phi = 0 \\
&\quad \text{or } 4r^2 \cos^2 \phi - 2r \cos \phi - (1 + r^2) = 0 \quad (**)
\end{aligned}$$

Now $r \sin \phi = 0 \implies z \in \mathbb{R}$. So for a solution to exist that corresponds to $z \in C_r \setminus \mathbb{R}$, (**) (which is quadratic in $\cos \phi$) must have a root with $|\cos \phi| < 1$ (since $\cos \phi = \pm 1$ corresponds to $z \in \mathbb{R}$). Using the quadratic formula, the roots of (**) are given by $\cos(\phi) = \frac{1 \pm \sqrt{5+4r^2}}{4r}$. Then (recalling that $r > 0$) we see that

$$\begin{aligned}
\left| \frac{1 - \sqrt{5+4r^2}}{4r} \right| < 1 &\iff 5 + 4r^2 < (4r + 1)^2 \\
&\iff 12(r + 1) \left(r - \frac{1}{3} \right) > 0 \\
&\iff r > \frac{1}{3}
\end{aligned}$$

Hence we see that $r_1^{(h)} = \frac{1}{3}$. At $r = r_1^{(h)}$, there are no self-intersections as the solution to (**) corresponds to $z \in \mathbb{R}$. For $r > r_1^{(h)}$, we see that there is (at least) one self-intersection at $w = h(re^{i\phi_0}) = h(re^{-i\phi_0})$, where $\phi_0 \in (0, \pi)$ satisfies $\cos \phi_0 = \frac{1 - \sqrt{5+4r^2}}{4r}$.

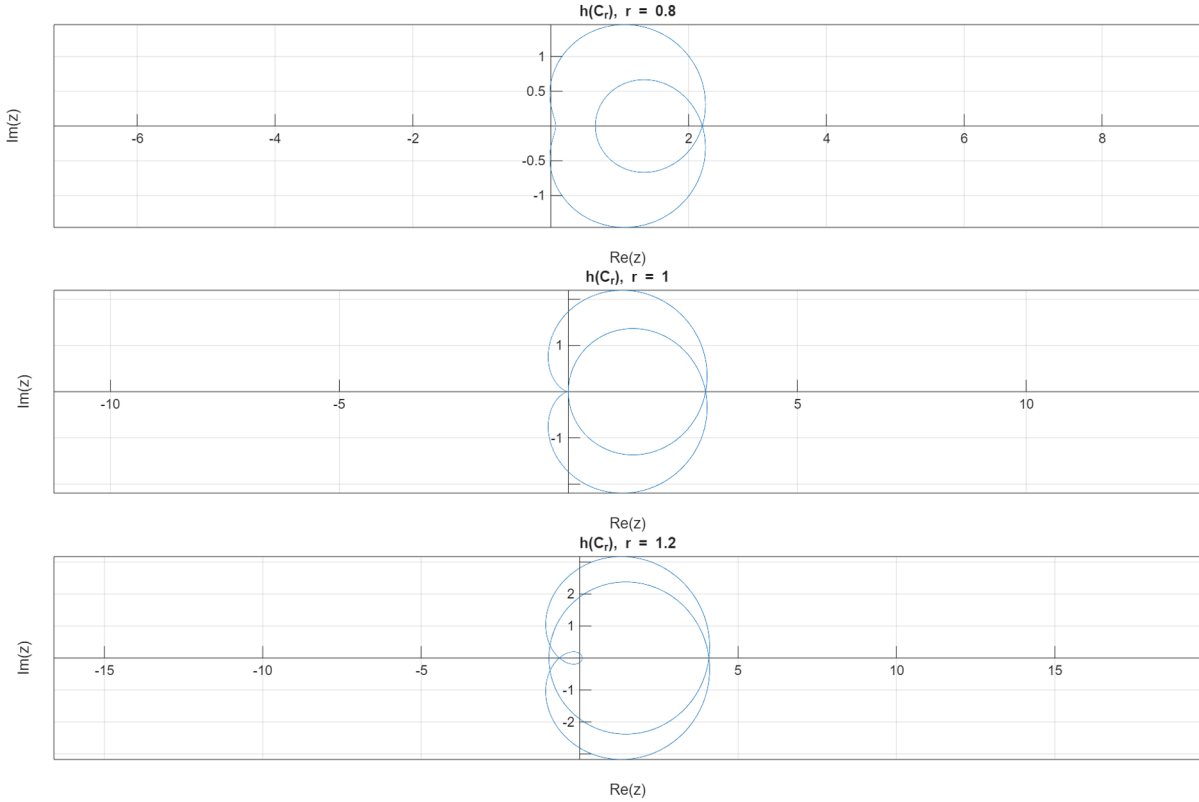


Figure 9: $h(C_r)$ for $r = 0.8$, $r = 1$, $r = 1.2$

As r increases beyond $r = r_1^{(h)}$, the interior loop on the right continues to increase in size, while the left side of the outer loop deforms inwards. At $r = r_2^{(h)} = 1$, the two loops meet, resulting in a self-intersection at the origin. For $r > r_2^{(h)}$, we see an additional self-intersection along the real axis, forming an interior loop on the right, and two additional self-intersections where the right loop intersects the outer loop.

We see that the additional self-intersection along the real axis arises when $(**)$ has two distinct solutions $\cos \phi_0, \cos \phi_1 \in (-1, +1)$. Thus (recalling that $r > 0$) we have

$$\begin{aligned} \left| \frac{1 + \sqrt{5 + 4r^2}}{4r} \right| < 1 &\iff 5 + 4r^2 < (4r - 1)^2 \\ &\iff 12(r - 1)\left(r + \frac{1}{3}\right) > 0 \\ &\iff r > 1 \end{aligned}$$

Thus $r_2^{(h)} = 1$, as required. For $r = r_2^{(h)}$, we observe that the intersection of the left and right loops at the origin arises because $h(-1) = h(1) = 0$, rather than because $h(z) = h(\bar{z})$ for some $z \in C_r \setminus \mathbb{R}$.

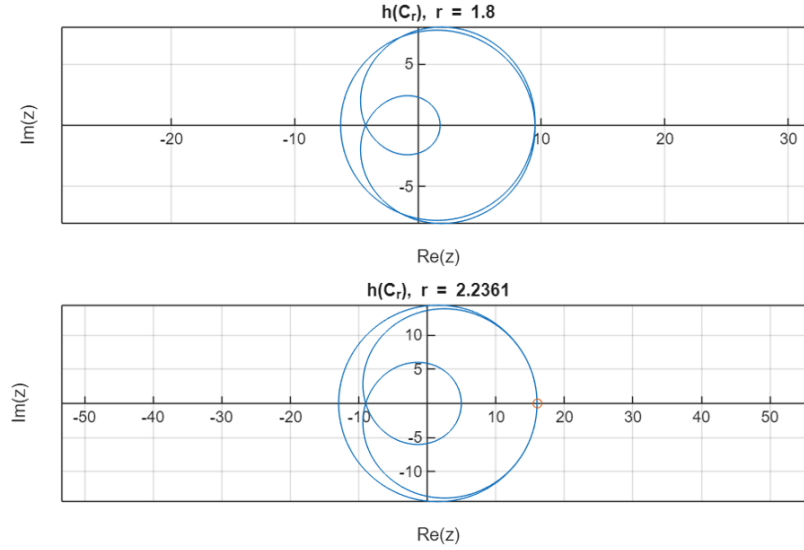


Figure 10: $h(C_r)$ for $r = 1.8$, $r = \sqrt{5}$

As r increases beyond $r = r_2^{(h)}$, we observe that the right loop continues to increase in size relative to the original loop. For $r \geq r_3^{(h)} \approx 2.24$, the right loop completely envelopes the original loop, and we no longer have the two self-intersections at $w, \bar{w} \in h(C_r) \setminus \mathbb{R}$.

To solve for $r_3^{(h)}$, we note that at $r = r_3^{(h)}$, the two loops must be tangent to one another at the self-intersection along the positive real-axis, circled in red in Figure (10). Recall that this lies at $w = h(z_\pm)$, where $z_\pm = re^{\pm i\phi_0}$ and $\phi_0 \in (0, \pi)$ satisfies $\cos \phi_0 = \frac{1 - \sqrt{5 + 4r^2}}{4r}$.

Using the symmetry of the curve about the real axis, we see that the tangents at w must be vertical, which is true when $\text{Re} \left[\frac{dh}{d\phi} \right] = 0$ at $\phi = \pm\phi_0$. Now $z = r_3^{(h)} e^{i\phi}$, so $\frac{dz}{d\phi} = ir_3^{(h)} e^{i\phi} = iz$, and we have

$$\begin{aligned} \frac{dh}{d\phi} &= \frac{dh}{dz} \frac{dz}{d\phi} \\ &= i(3z^3 - 2z^2 - z) \end{aligned}$$

So we require that $3z_{\pm}^3 - z_{\pm}^2 - z_{\pm} \in \mathbb{R}$. But we know that $h(z_{\pm}) = z_{\pm}^3 - z_{\pm}^2 - z_{\pm} + 1 = w \in \mathbb{R}$, so

$$\begin{aligned}
& 3z_{\pm}^3 - 2z_{\pm}^2 - z_{\pm} \in \mathbb{R} \\
& \iff z^2 + 2z \in \mathbb{R} \\
& \iff \operatorname{Im} \left[\left(r_3^{(h)} \cos \phi_0 + i r_3^{(h)} \sin \phi_0 \right) \left(2 + r_3^{(h)} \cos \phi_0 + i r_3^{(h)} \sin \phi_0 \right) \right] = 0 \\
& \iff 2r_3^{(h)} \sin \phi_0 \left(1 + r_3^{(h)} \cos \phi_0 \right) = 0
\end{aligned}$$

Since $r_3^{(h)} \sin \phi_0 \neq 0$, we have

$$\begin{aligned}
& r_3^{(h)} \cos \phi_0 = -1 \\
& \implies 1 - \sqrt{5 + 4 \left(r_3^{(h)} \right)^2} = -4 \\
& \implies r_3^{(h)} = \sqrt{5}
\end{aligned}$$

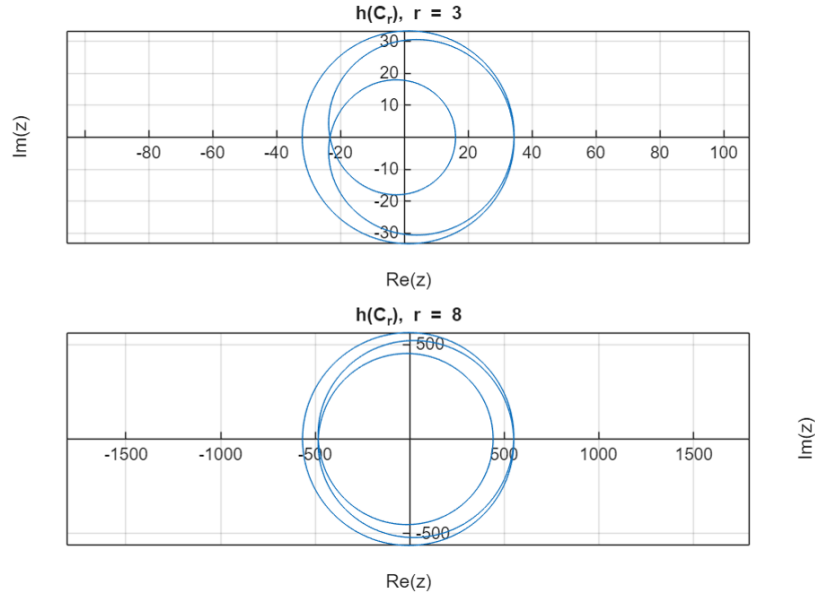


Figure 11: $h(C_r)$ for $r = 3, r = 8$

As r increases to ∞ , the three loops become approximately circular and approximately equal in size. For large r , the z^3 term in $h(z) = z^3 - z^2 - z + 1$ dominates. The image of C_r under $z \mapsto z^3$ is a circle of radius r^3 which winds around the origin thrice, agreeing with the limiting behaviour of $h(C_r)$ as $r \rightarrow \infty$.

Question 5.

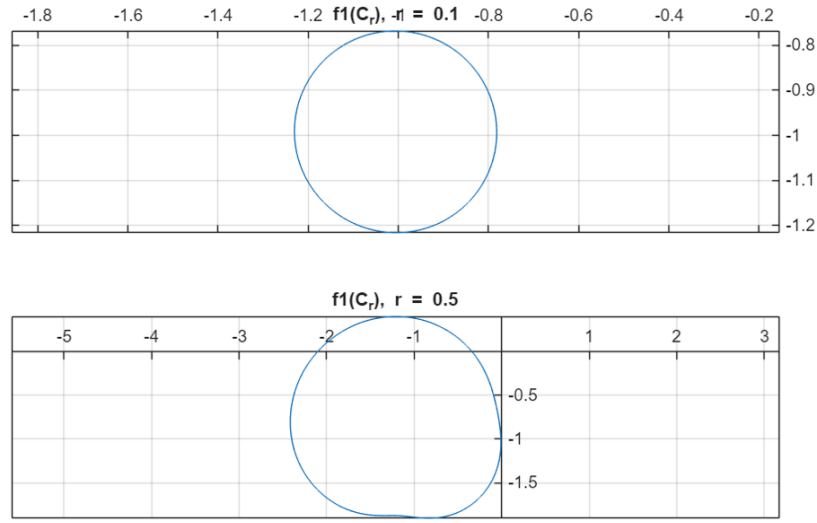


Figure 12: $f_1(C_r)$ for $r = 0.1$ and $r = 0.3$

As before, for small $r < 1$, the $(2 - i)z - 1 - i$ terms in $f_1(z) = z^3 - z^2 + (2 - i)z - 1 - i$ dominate, so $f_1(C_r)$ approximates a circle of radius $\sqrt{5}r$ centred on $-1 - i$.

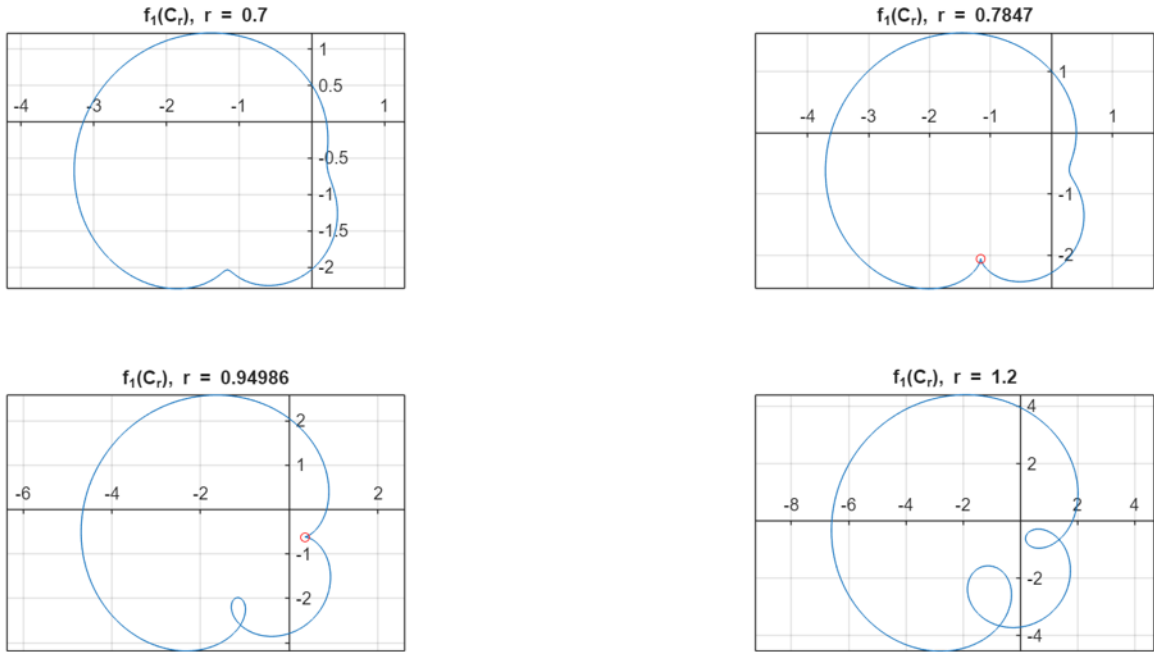


Figure 13: $f_1(C_r)$ for $r = 0.7$, $r = \rho_1$, $r = \rho_2$, $r = 1.2$

As r increases from $r = 0$, we see that $f_1(C_r)$ remains a simple curve as the right and bottom sides deform inwards, up until $r = r_1^{(f_1)} \approx 0.78$. For $r > r_1^{(f_1)}$, we see a self-intersection at the bottom, and an

additional loop forms inside. For $r > r_2^{(f_1)} \approx 0.95$, we see another self-intersection at the right, and another additional loop forms inside.

To solve for $r_1^{(f_1)}, r_2^{(f_1)}$, we first note that the tangent vector $\mathbf{t}(s)$ (as defined in Section (4)) is undefined at the points circled in red in Figure (13). Note (as we will prove in Question (6)) that $\mathbf{t}(s) = \frac{\mathbf{x}'(\phi)}{|\mathbf{x}'(\phi)|}$. Thus we must have $\frac{df_1}{d\phi} = 0$ at the highlighted points. But

$$\begin{aligned} \frac{df_1}{d\phi} = 0 &\iff \frac{df_1}{dz} \frac{dz}{d\phi} = 0 \\ &\iff izf_1'(z) = 0 \end{aligned}$$

We know $z \neq 0$ since $r > 0$. From Question 2, we know the roots of w_1, w_2 of f_1' . Then we must have $r_1^{(f_1)} = |w_1| = \rho_1, r_2^{(f_1)} = |w_2| = \rho_2$ where the values of ρ_1, ρ_2 are given by Table (2). From the exact form of the roots, algebraic expressions for $r_1^{(f_1)}, r_2^{(f_1)}$ are

$$\begin{aligned} r_1^{(f_1)} &= \left[\left(\frac{1}{3} - \sqrt{\frac{\sqrt{34}-5}{18}} \right)^2 + \frac{\sqrt{34}+5}{18} \right]^{1/2} \\ r_2^{(f_1)} &= \left[\left(\frac{1}{3} + \sqrt{\frac{\sqrt{34}-5}{18}} \right)^2 + \frac{\sqrt{34}+5}{18} \right]^{1/2} \end{aligned}$$

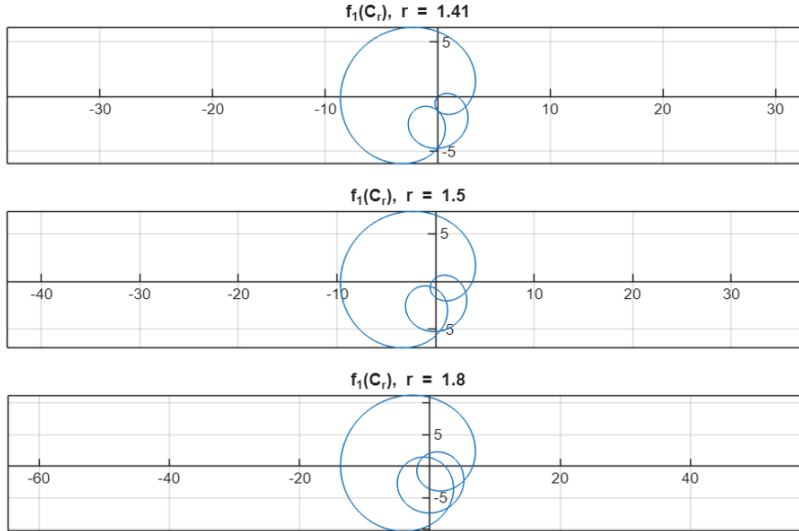


Figure 14: $f_1(C_r)$ for $r = 1.41, r = 1.5, r = 1.8$

As r increases beyond $r = r_2^{(f_1)}$, we observe that the two interior loops increase in size relative to the original loop, while the self-intersections at the base of the bottom loop moves towards the right. For $r \geq r_3^{(f_1)} \approx 1.41$, the two interior loops intersect. When $r = r_3^{(f_1)}$, this results in one additional self-intersection, while for $r > r_3^{(f_1)}$, this results in two additional self-intersections.

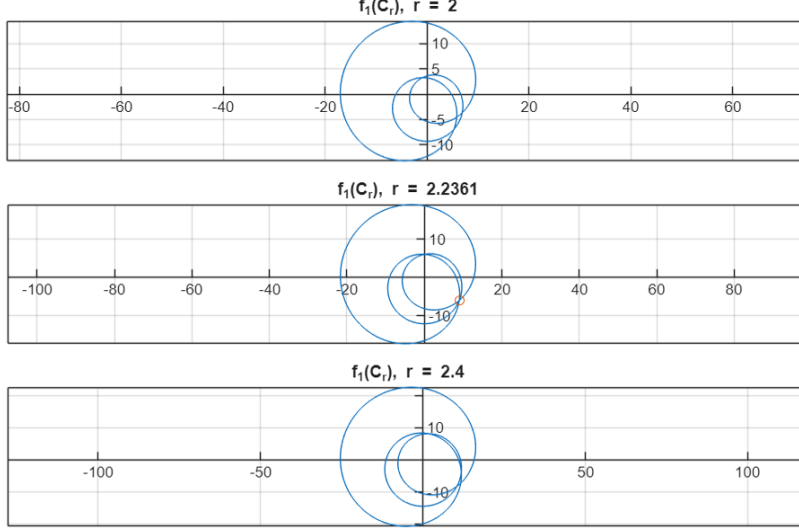


Figure 15: $f_1(C_r)$ for $r = 2.0$, $r = \sqrt{5}$, $r = 2.4$

As r increases beyond $r = r_3^{(f_1)}$, the two interior loops continue to increase in size while the self-intersections at the bases of the loops move towards each other. For $r = r_4^{(f_1)} \approx 2.24$, the bases of the loops coincide, so $f_1(C_r)$ has exactly two self-intersections (and at one of the intersections, $f_1(C_r)$ passes through the point three times). For $r > r_4^{(f_1)}$, the curve $f_1(C_r)$ has four self-intersections, as before.

To find $r_4^{(f_1)}$, we observe that when $r = r_4^{(f_1)}$, the curve $f_1(C_r)$ passes through the point w (circled in red in Figure (15)) thrice. Then $f_1(z) - w = z^3 - z^2 + (2 - i)z - 1 - i - w$ has three roots in C_r .

Observe that when $w = 9 - 6i$,

$$\begin{aligned} f_1(z) - w &= z^3 - z^2 + (2 - i)z - 10 + 5i \\ &= (z + 1 + 2i)(z^2 - (2 + 2i)z + 5i) \end{aligned}$$

Using the quadratic formula, this has roots

$$\begin{aligned} z_1 &= -1 - 2i \\ z_2 &= \left(1 + \sqrt{\frac{3}{2}}\right) + \left(1 - \sqrt{\frac{3}{2}}\right)i \\ z_3 &= \left(1 - \sqrt{\frac{3}{2}}\right) + \left(1 + \sqrt{\frac{3}{2}}\right)i \end{aligned}$$

These satisfy $|z_1| = |z_2| = |z_3| = \sqrt{5}$. Then $r_4^{(f_1)} = \sqrt{5}$.

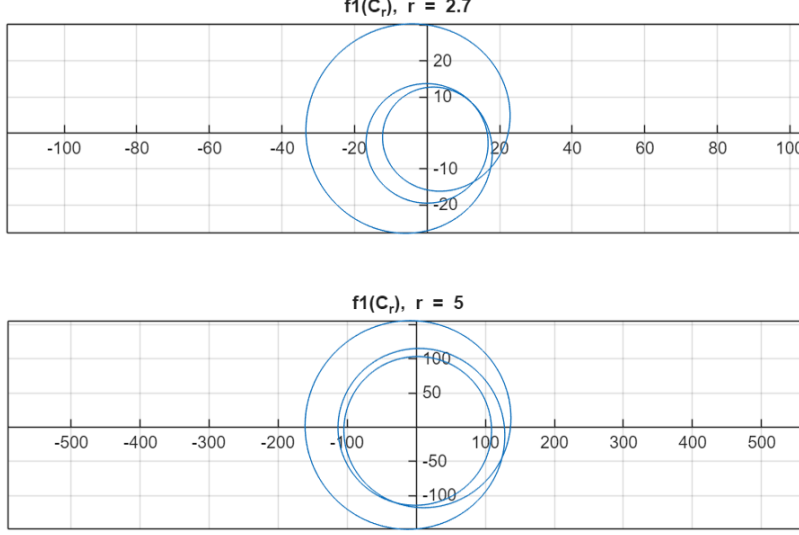


Figure 16: $f_1(C_r)$ for $r = 2.7, r = 5.0$

As r increases beyond $r = r_4^{(f_1)}$, the two interior loops stop intersecting at $r = r_5^{(f_1)} \approx 2.65$. Beyond this value, as r increases, the z^3 term in $f_1(z) = z^3 - z^3 + (2-i)z - 1 - i$ dominates, and the three loops become approximately circular, as in the cases of $g(C_r)$ and $h(C_r)$.

4 Curvature of images $f(C_r)$ of C_r

Question 6.

We write $\dot{\mathbf{x}} = \frac{d\mathbf{x}}{ds}$ and $\mathbf{x}' = \frac{d\mathbf{x}}{d\phi}$, and similarly for the scalars x and y . We parameterise C_r by $z(\phi) = re^{i\phi}$ and observe that $\frac{ds}{d\phi} \geq 0$ for all $\phi \in [0, 2\pi]$. We have

$$\begin{aligned}\dot{\mathbf{x}} &= (\dot{x}, \dot{y}) \\ &= (x', y') \frac{d\phi}{ds} \\ \ddot{\mathbf{x}} &= \frac{d}{ds} \left[(x', y') \frac{d\phi}{ds} \right] \\ &= (x'', y'') \left[\frac{d\phi}{ds} \right]^2 + (x', y') \left[\frac{d^2\phi}{ds^2} \right]\end{aligned}$$

Now, by the chain rule,

$$\begin{aligned}\mathbf{x}' \frac{d\phi}{ds} &= \dot{\mathbf{x}} = \mathbf{t} \text{ is a unit vector} \\ \therefore \frac{d\phi}{ds} &= \frac{|\dot{\mathbf{x}}|}{|\mathbf{x}'|} = \frac{1}{|\mathbf{x}'|} \\ \therefore \frac{d^2\phi}{ds^2} &= \frac{d}{d\phi} \left[\frac{1}{|\mathbf{x}'|} \right] \frac{d\phi}{ds} \\ &= -\frac{1}{2|\mathbf{x}'|^3} \frac{d}{d\phi} [|\mathbf{x}'|^2] \frac{d\phi}{ds} \\ &= -\frac{\mathbf{x}' \cdot \mathbf{x}''}{|\mathbf{x}'|^4}\end{aligned}$$

Thus, noting that $|\mathbf{x}'|^2 = x'^2 + y'^2$ and $\mathbf{x}' \cdot \mathbf{x}'' = x'x'' + y'y''$, we have

$$\begin{aligned}
\ddot{\mathbf{x}} &= \frac{1}{|\mathbf{x}'|^4} (|\mathbf{x}'|^2 x'' - \mathbf{x}' \cdot \mathbf{x}'' x', |\mathbf{x}'|^2 y'' - \mathbf{x}' \cdot \mathbf{x}'' y') \\
|\ddot{\mathbf{x}}|^2 &= \frac{1}{|\mathbf{x}'|^8} [|\mathbf{x}'|^4 (x''^2 + y''^2) - 2|\mathbf{x}'|^2 \mathbf{x}' \cdot \mathbf{x}'' (x'x'' + y'y'') + (\mathbf{x}' \cdot \mathbf{x}'')^2 (x'^2 + y'^2)] \\
&= \frac{1}{|\mathbf{x}'|^6} [x'y'' - y'x'']^2 \\
&= \frac{|\mathbf{x}' \times \mathbf{x}''|^2}{|\mathbf{x}'|^6} \\
|\kappa| &= \frac{|\mathbf{x}' \times \mathbf{x}''|}{|\mathbf{x}'|^3} \quad \text{as required.}
\end{aligned}$$

We have

$$\begin{aligned}
\mathbf{x}' &= \dot{\mathbf{x}} \frac{ds}{d\phi} = \mathbf{t}(s) \frac{ds}{d\phi} \\
\mathbf{x}'' &= \frac{d}{d\phi} \left[\mathbf{t} \frac{ds}{d\phi} \right] = \dot{\mathbf{t}} \left[\frac{ds}{d\phi} \right]^2 + \mathbf{t} \left[\frac{d^2 s}{d\phi^2} \right] \\
&= \mathbf{k}(s) \left[\frac{ds}{d\phi} \right]^2 + \mathbf{t}(s) \left[\frac{d^2 s}{d\phi^2} \right]
\end{aligned}$$

We can view $\mathbb{C}$ as the two-dimensional plane $\{z = 0\} \in \mathbb{R}^3$ under the embedding $x + iy \mapsto (x, y, 0)$. Geometrically, we see that the curve C_r is turning anticlockwise in $\mathbb{C}$ at a point $\mathbf{x}(s) \in C_r$ if and only if the z -component of $\mathbf{t}(s) \times \mathbf{k}(s)$ is positive. Denote by $\mathbf{u} \wedge \mathbf{v} = u_1 v_2 - u_2 v_1$ the z -component of $\mathbf{u} \times \mathbf{v}$.

Then, recalling that $\frac{ds}{d\phi} \geq 0$ and that $\frac{ds}{d\phi} = 0 \implies \mathbf{x}' = \mathbf{0} \implies \kappa = 0$, and observing that $\mathbf{t}(s) \wedge \mathbf{t}(s) = 0$, we see that

$$\begin{aligned}
\operatorname{sgn} \kappa(s) &= \operatorname{sgn} [\mathbf{t}(s) \wedge \mathbf{k}(s)] \\
&= \operatorname{sgn} \left[\left(\mathbf{t}(s) \frac{ds}{d\phi} \right) \wedge \left(\mathbf{k}(s) \left[\frac{ds}{d\phi} \right]^2 + \mathbf{t}(s) \left[\frac{d^2 s}{d\phi^2} \right] \right) \right] \\
&= \operatorname{sgn} [\mathbf{x}' \wedge \mathbf{x}'']
\end{aligned}$$

Recall that we have $z = re^{i\phi}$. Then $\frac{dz}{d\phi} = ire^{i\phi} = iz$. Thus we have

$$\begin{aligned}
x + iy &= f(z) \\
x' + iy' &= \frac{df}{dz} \frac{dz}{d\phi} = iz \frac{df}{dz} \\
\therefore \mathbf{x}' &= \left(-\operatorname{Im} \left[z \frac{df}{dz} \right], \operatorname{Re} \left[z \frac{df}{dz} \right] \right) \\
x'' + iy'' &= \frac{d}{dz} \left[iz \frac{df}{dz} \right] \frac{dz}{d\phi} = iz \left[i \frac{df}{dz} + iz \frac{d^2 f}{dz^2} \right] \\
\therefore \mathbf{x}'' &= \left(\operatorname{Re} \left[-z \frac{df}{dz} - z^2 \frac{d^2 f}{dz^2} \right], \operatorname{Im} \left[-z \frac{df}{dz} - z^2 \frac{d^2 f}{dz^2} \right] \right)
\end{aligned}$$

We observe that $\mathbf{u} \wedge \mathbf{v} = \text{Im}(\bar{u}v)$. Thus we have

$$\begin{aligned}\kappa &= \frac{\mathbf{x}' \wedge \mathbf{x}''}{|\mathbf{x}'|^3} = \frac{\text{Im} \left[\left(i z \frac{df}{dz} \right) \left(-z \frac{df}{dz} - z^2 \frac{d^2 f}{dz^2} \right) \right]}{\left| i z \frac{df}{dz} \right|^3} \\ &= \frac{\text{Re} \left[\left(z \frac{df}{dz} \right) \left(z \frac{df}{dz} + z^2 \frac{d^2 f}{dz^2} \right) \right]}{\left| z \frac{df}{dz} \right|^3} \\ \int_{f(C_r)} \kappa \, ds &= \int_{\phi=0}^{2\pi} \kappa \frac{ds}{d\phi} \, d\phi = \int_{\phi=0}^{2\pi} \frac{\text{Re} \left[\left(z \frac{df}{dz} \right) \left(z \frac{df}{dz} + z^2 \frac{d^2 f}{dz^2} \right) \right]}{\left| z \frac{df}{dz} \right|^3} |\mathbf{x}'| \, d\phi \\ &= \int_{\phi=0}^{2\pi} \left[1 + \frac{\text{Re} \left[\left(\frac{df}{dz} \right) \left(z \frac{d^2 f}{dz^2} \right) \right]}{\left| \frac{df}{dz} \right|^2} \right] d\phi\end{aligned}$$

We observe that, traversing the curve C_r counterclockwise (in the direction of increasing ϕ), the image curves $f_1(C_r), g(C_r), h(C_r)$ turn anticlockwise about the local centre of rotation over most of the curve. For examples where the curve turns clockwise, refer to Figures (7) and (13); we see that the curve is turning clockwise about the local centre of rotation near where $h(C_r), r = 0.3$ intersects the real axis on the right, and similarly at the bottom and right sides of $f_1(C_r), r = 0.7$. To verify that our program calculates the sign of κ correctly, we plot $\kappa(\phi)$ for selected values of r .

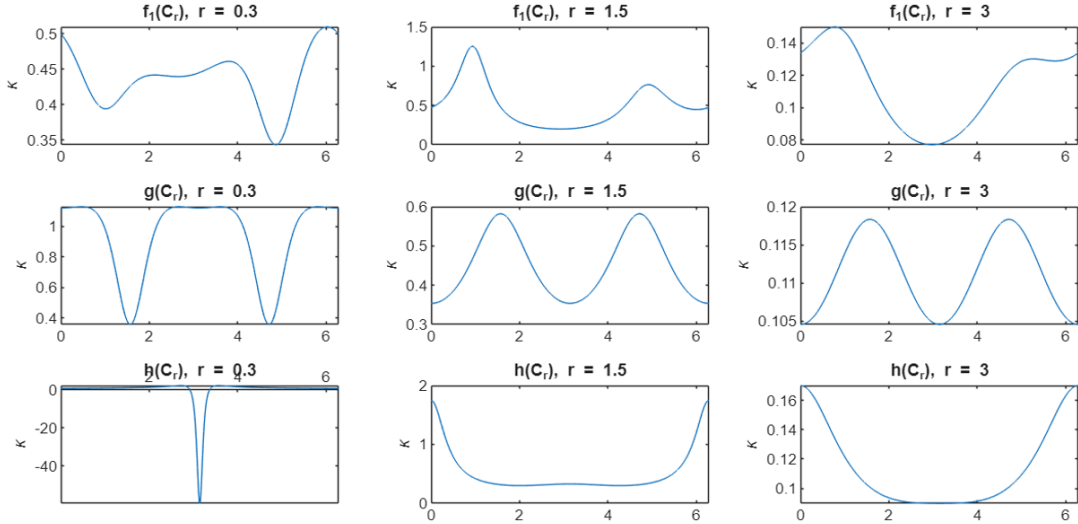


Figure 17: κ against ϕ for $f_1(C_r), g(C_r), h(C_r)$, $r = 0.3, 1.5, 3.0$

In line with our observations, the program returns positive values for most values of ϕ , but is negative for a small range of values of ϕ for $h(C_r), r = 0.3$.

Question 7.

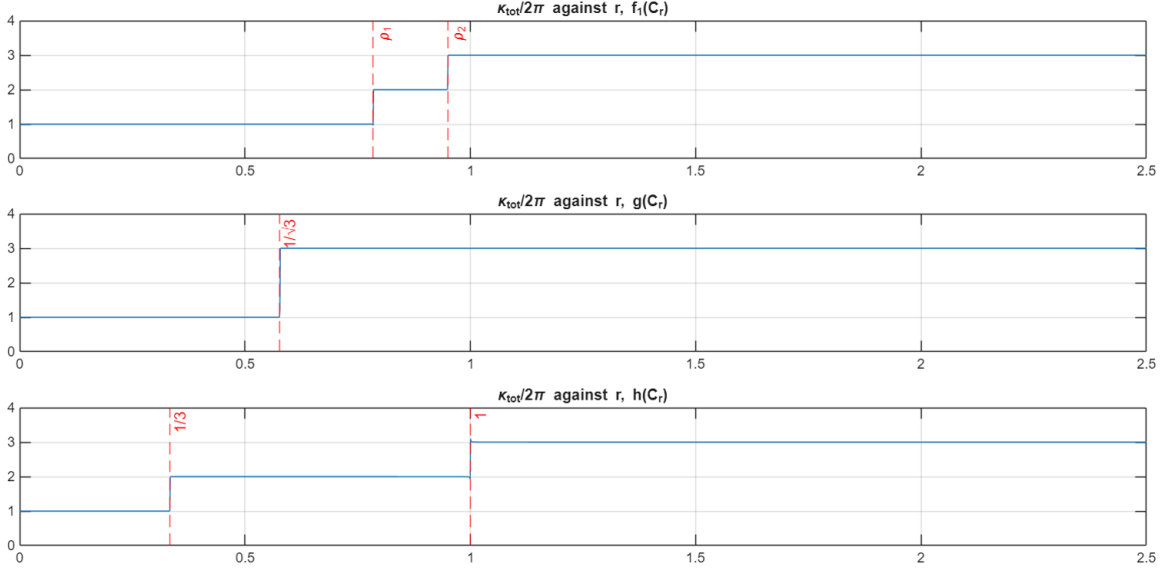


Figure 18: $\frac{\kappa_{\text{tot}}}{2\pi}$ against r for $f_1(C_r)$, $g(C_r)$, $h(C_r)$, and $r \in [0, 2.5]$

From Figure (18), we observe that $\kappa_{\text{tot}}(f; r)$ appears to take on discrete values in $\{2n\pi\}_{n \in \mathbb{Z}}$.

For $f_1(C_r)$, $\kappa_{\text{tot}}(f_1; r)$ jumps from 2π to 4π at $r = \rho_1$, and from 4π to 6π at $r = \rho_2$. Referring to our analysis in Question 5, we see that this corresponds to the formation of an additional loop at $r = r_1^{(f_1)}$ and at $r = r_2^{(f_1)}$ respectively. We can use our program to compute κ_{tot} at these values of r and find that, to three significant figures, $\frac{\kappa_{\text{tot}}(f_1, \rho_1)}{2\pi} = 1.50$ and $\frac{\kappa_{\text{tot}}(f_1, \rho_2)}{2\pi} = 2.50$.

For $g(C_r)$, $\kappa_{\text{tot}}(g; r)$ jumps from 2π to 6π at $r = \frac{1}{\sqrt{3}}$. Referring to our analysis in Question 3, we see that this corresponds to the formation of two additional loops at $r = r_1^{(g)}$. Again, we can compute $\frac{\kappa_{\text{tot}}(g, \frac{1}{\sqrt{3}})}{2\pi} = 2.00$ (to three significant figures).

For $f_1(C_r)$, $\kappa_{\text{tot}}(h; r)$ jumps from 2π to 4π at $r = \frac{1}{3}$, and from 4π to 6π at $r = 1$. Referring to our analysis in Question 4, we see that this corresponds to the formation of an additional loop at $r = r_1^{(h)}$ and at $r = r_2^{(h)}$ respectively. We find that $\frac{\kappa_{\text{tot}}(h, \frac{1}{3})}{2\pi} = 1.50$ and $\frac{\kappa_{\text{tot}}(h, 1)}{2\pi} = 2.50$ to three significant figures.

Therefore, for the three polynomials investigated, $\frac{\kappa_{\text{tot}}(f; r)}{2\pi}$ represents the number of loops in the curve $f(C_r)$ at the given value of r , except at a finite number of points which are discontinuities of $\frac{\kappa_{\text{tot}}(f; r)}{2\pi}$, where $\frac{\kappa_{\text{tot}}(f; r)}{2\pi}$ takes on the average value of points on either side of the discontinuity. We hypothesise that this holds for a general polynomial f .

Let $\mathbf{x}(s)$ be a curve with tangent $\mathbf{t}(s) = \dot{\mathbf{x}}(s)$ everywhere nonsingular. Recall that $\mathbf{t}(s)$ always lies on the unit circle. We define the **rotation index** of $\mathbf{x}(s)$ to be the winding number of $\mathbf{t}(s)$ about the origin, and we see that this is equivalent to what we previously informally referred to as the ‘number of loops’ of the curve $\mathbf{x}(s)$.

Let f be a complex polynomial. Recalling that $\frac{dz}{d\phi} = iz$, we see that the curve $f(C_r)$ has tangent given by

$$t(\phi) = \frac{\frac{df}{d\phi}}{\left| \frac{df}{d\phi} \right|} = \frac{iz \frac{df}{dz}}{r \left| \frac{df}{dz} \right|}$$

Its derivative with respect to ϕ is then

$$\frac{dt}{d\phi} = \frac{i}{r} \frac{d}{d\phi} \left(\frac{z \frac{df}{dz}}{\left| \frac{df}{dz} \right|} \right) = \frac{i}{r} \left[\frac{\frac{df}{dz} + z \frac{d^2 f}{dz^2}}{\left| \frac{df}{dz} \right|} \frac{dz}{d\phi} - \frac{z \frac{df}{dz}}{2 \left| \frac{df}{dz} \right|^3} \frac{d}{d\phi} \left(\left| \frac{df}{dz} \right|^2 \right) \right]$$

Now

$$\begin{aligned} \frac{d}{d\phi} \left(\left| \frac{df}{dz} \right|^2 \right) &= \frac{d}{d\phi} \left(\frac{df}{dz} \overline{\frac{df}{dz}} \right) = \frac{df}{dz} \frac{d}{d\phi} \left(\overline{\frac{df}{dz}} \right) + \overline{\frac{df}{dz}} \frac{d}{d\phi} \left(\frac{df}{dz} \right) \\ &= \frac{df}{dz} \frac{d}{d\phi} \left(\operatorname{Re} \left[\frac{df}{dz} \right] - \operatorname{Im} \left[\frac{df}{dz} \right] \right) + \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2} \frac{dz}{d\phi} \\ &= \frac{df}{dz} \left(\operatorname{Re} \left[\frac{d}{d\phi} \frac{df}{dz} \right] - \operatorname{Im} \left[\frac{d}{d\phi} \frac{df}{dz} \right] \right) + iz \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2} \\ &= \frac{df}{dz} \overline{\frac{d}{d\phi} \left(\frac{df}{dz} \right)} + iz \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2} \\ &= -i \bar{z} \frac{df}{dz} \overline{\frac{d^2 f}{dz^2}} + iz \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2} \end{aligned}$$

Thus

$$\frac{dt}{d\phi} = -\frac{z}{r} \left[\frac{\frac{df}{dz} + z \frac{d^2 f}{dz^2}}{\left| \frac{df}{dz} \right|} + \frac{\frac{df}{dz} \left(\bar{z} \frac{df}{dz} \overline{\frac{d^2 f}{dz^2}} - z \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2} \right)}{2 \left| \frac{df}{dz} \right|^3} \right]$$

We see that $t : [0, 2\pi] \mapsto \mathbb{C}$ defines a closed curve γ . From IB Complex Analysis, the winding number of this curve about the origin is given by

$$\begin{aligned} I(\gamma; 0) &= \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z} dz \\ &= \frac{1}{2\pi i} \int_{\phi=0}^{2\pi} \frac{1}{t(\phi)} \frac{dt}{d\phi} d\phi \\ &= \frac{1}{2\pi} \int_{\phi=0}^{2\pi} \left[1 + \frac{z \frac{d^2 f}{dz^2}}{\frac{df}{dz}} + \frac{\bar{z} \frac{df}{dz} \overline{\frac{d^2 f}{dz^2}} - z \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2}}{2 \left| \frac{df}{dz} \right|^2} \right] d\phi \\ &= \frac{1}{2\pi} \int_{\phi=0}^{2\pi} \left[1 + \frac{\bar{z} \frac{df}{dz} \overline{\frac{d^2 f}{dz^2}} + z \overline{\frac{df}{dz}} \frac{d^2 f}{dz^2}}{2 \left| \frac{df}{dz} \right|^2} \right] d\phi \\ &= \frac{1}{2\pi} \int_{\phi=0}^{2\pi} \left[1 + \frac{\operatorname{Re} \left[\left(\frac{df}{dz} \right) \left(z \frac{d^2 f}{dz^2} \right) \right]}{\left| \frac{df}{dz} \right|^2} \right] d\phi \\ &= \frac{\kappa_{\text{tot}}}{2\pi}, \text{ as hypothesised.} \end{aligned}$$

5 Program Listings

Plotting $f(C_r)$ and finding the point closest to the origin

```
%Programming Task 1
r = input("r = "); %Input r value

t = linspace(0, 2*pi, 5000);
z = r*exp(1i*t); %2000 evenly spaced points on C_r
w = f1(z); %

[M, I] = min(abs(w));
disp(['nearest point = ' num2str(w(I)) ', distance = ' num2str(M)])

plot(w, "-")
hold on
plot(w(I), '.', 'MarkerSize', 20)
axis equal
ax = gca;
ax.XAxisLocation = 'origin';
ax.YAxisLocation = 'origin';
grid on
xlabel("Re(z)")
ylabel("Im(z)")
title("f(C_r)")
hold off
```

Root finder

```
grain = 25000; %Number of values of r investigated
r = linspace(1.55, 1.65, grain); %Values of r investigated

x = zeros(grain,1);

for i = 1:grain
    x(i) = f1(pt1func(r(i))); %arg min of f_1(z) for z in C_r, for a given r
end

plot(r, abs(x), "-")
xlabel("r")
ylabel("min(|f_1(C_r)|)")
[M, I] = min(abs(x)); %arg min of f_1(z) for z in C_r, over all r in the range
disp([num2str(M) ', ' num2str(x(I)) ', ' num2str(r(I), 6) ', ' num2str(pt1func(r(I)), 6)])
```

Computing κ_{tot}

```
grain = 500; %Number of values of r investigated
r = linspace(0, 3, grain); %Values of r investigated
int = zeros(grain,1);

for count = 1:grain
    t = linspace(0, 2*pi, 5000);
    z = r(count)*exp(1i*t); %5000 points in C_r
    for x = z %Approximating the integral by a finite sum over points in C_r
        int(count) = int(count) + (1 + real(conj(gp(x)) .* x .* gpp(x)) ./ ( abs(gp(x)).^2 ) .* (2*pi / 5000));
    end
end

plot(r, int)|
```

Plotting $\kappa(\phi)$

```
for r = [0.3, 1.5, 3.0]
    nexttile
    t = linspace(0, 2*pi, 5000);
    z = r*exp(1i*t); %5000 points in C_r
    kappa = (1 + real(conj(f1p(z)) .* z .* f1pp(z)) ./ ( abs(f1p(z)).^2 ) ) ./ abs(f1p(z));
    %Calculate kappa at each of the 5000 points
    plot(t, kappa)
    ax = gca;
    ax.XAxisLocation = 'origin';
    ax.YAxisLocation = 'origin';
    ylabel("\kappa")
    title("f_1(C_r), r = " + r)
end
```